DYNAMIC PORTFOLIO OPTIMIZATION WITH GRADIENT BOOSTING ENSEMBLES

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ABSTRACT

This study develops a dynamic asset allocation strategy for the "Hull Tactical Market Prediction" Kaggle competition, utilizing gradient boosting models (LightGBM and XGBoost) to examine the practical limits of the Efficient Market Hypothesis (EMH). By integrating historical risk metrics with model signals, we create a volatility-aware portfolio weighting framework. The strategy achieves an adjusted Sharpe ratio of 0.7138 on validation data and generalizes effectively, demonstrating machine learning's ability to identify and exploit short-term market inefficiencies under disciplined risk control.

1 INTRODUCTION

1.1 RESEARCH BACKGROUND AND MOTIVATION

The Efficient Market Hypothesis (EMH) has long been regarded as the cornerstone of financial theory, asserting that asset prices fully reflect all available information, making it impossible for investors to consistently achieve excess returns through market timing (Fama, 1970). However, behavioral finance points to numerous market "anomalies," such as excessive volatility and price bubbles, challenging the assumption of perfect rationality (Shiller, 2003).

In parallel, the rise of big data and machine learning has introduced new tools for testing this theory. Modern algorithms, particularly gradient boosting decision trees such as XGBoost (Chen & Guestrin, 2016) and LightGBM (Ke et al., 2017), can uncover subtle nonlinear patterns in high-dimensional financial data that traditional models may miss. This capability lies at the core of contemporary empirical asset pricing research (Gu et al., 2020) and opens new avenues for tactical asset allocation.

Industry practice has already led the way in this exploration. For example, the Hull Tactical Fund builds predictive models by integrating multi-source data—including technical indicators, economic data, and social media sentiment—with strong back test performance providing empirical evidence of capturing short-lived market inefficiencies.

Against this backdrop, the "Hull Tactical Market Prediction" Kaggle competition serves as an ideal test bed connecting theory, practice, and technology. It requires participants to predict S&P 500 excess returns and adjust portfolios dynamically. This project aims to participate in the competition, focusing on a comparative implementation of the advanced XGBoost and LightGBM frameworks, to systematically investigate whether and how quantitative models can effectively identify and exploit short-term predictive signals in today's complex data environment—thereby offering new empirical insights into the practical boundaries of the Efficient Market Hypothesis.

1.2 PROBLEM DEFINITION

Formally, the competition requires participants to predict, for each trading day in the test period, a portfolio weight $w_t \in [0, 2]$, which represents the proportion of capital allocated to the S&P 500 index. A weight of 0 corresponds to holding cash, 1 indicates full investment, and 2 represents a

200% leveraged long position. The objective is to maximize a volatility-adjusted Sharpe ratio – a customized evaluation metric that rewards excess returns while penalizing strategies whose volatility significantly exceeds that of the market or whose performance falls short of the market benchmark.

This formulation transforms the problem from a pure regression task (predicting returns) into a structured decision-making problem, where predictions must be directly translated into actionable portfolio weights to optimize risk-adjusted performance.

1.3 Research Methodology and Technical Approach

This project adopts the following technical approach:

1. Data Preprocessing Stage: Handling missing values, outliers, and feature standardization.

2. Feature Engineering Stage: Constructing lagged features, rolling statistics, and technical indicators.

3. Model Building Stage: We focus on implementing two gradient boosting tree models, XGBoost and LightGBM. Cross-validation results indicate that LightGBM slightly outperforms XGBoost (RMSE: 0.007383 vs. 0.007628).

4. Strategy Integration Stage: Converting predicted values into investment weights that satisfy the given constraints.

5. Evaluation and Optimization Stage: Using time-series cross-validation and custom evaluation metrics for validation and optimization.

2 Data exploration & feature engineering

The training dataset consists of 9,021 data and 98 features. To ensure model reliability and avoid noise introduced by data imperfections, we conducted a systematic data cleaning and feature refinement process, detailed as follows.

2.1 Missing Value Analysis

Our initial inspection revealed that 12 features contained more than 3,000 missing entries. Such extensive sparsity renders imputation methods (e.g., mean or median filling) unreliable, as they would severely distort the true distribution of these variables. Therefore, we removed these 12 features entirely from the dataset.

For the remaining features, we evaluated missingness at the row level. A total of 1,006 rows exhibited more than 30 missing feature values, indicating substantial information loss. Retaining these rows would introduce noise and bias into the model, hence we removed these 1,006 observations during preprocessing.

2.2 Correlation Analysis

To understand the interdependence among features, we constructed a correlation heatmap to inspect pairwise relationships (see Figure 1). The analysis identified nine groups of features with correlation coefficients exceeding 0.85. Highly correlated predictors provide limited incremental information to our models and may increase training time while adding little predictive value.

Thus, to improve feature diversity and reduce redundancy, we discarded one feature from each high-correlation group, keeping only the representative variable with the strongest individual relevance.

2.3 Imputation

After removing highly sparse features and heavily incomplete rows, the remaining features exhibited relatively minor missingness. For these variables, we implemented a rolling-window imputation scheme, replacing missing values using the average of the preceding and following 250 trading

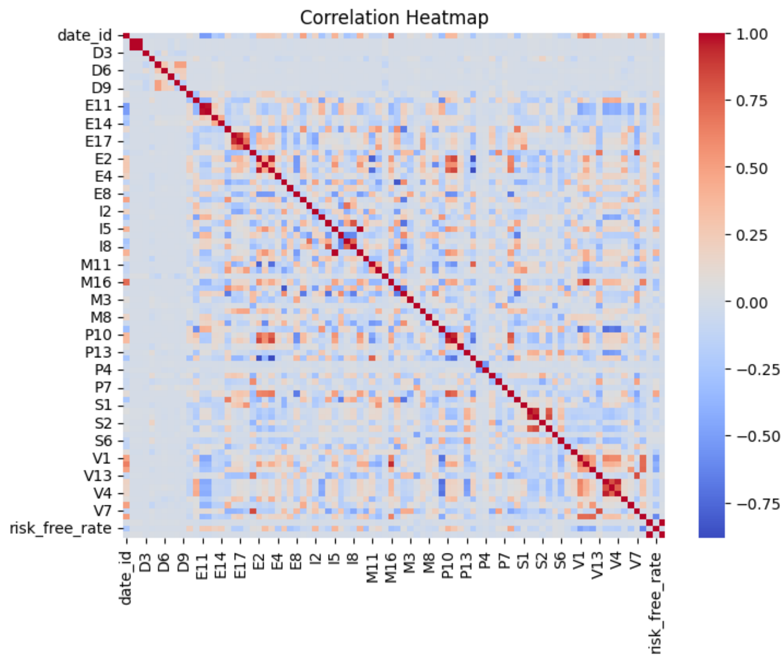

Figure 1: Correlation heatmap of features.

days. This method preserves the temporal structure of the data and avoids forward-looking bias, while producing smoother and more realistic imputations than static summary statistics.

2.4 Final Cleaned Dataset

Our cleaned dataset contains 8,015 valid data and 76 features. This cleaned dataset provides a robust foundation for downstream model training.

3 Model Selection & Training

We employed an ensemble learning approach combining two state-of-the-art gradient boosting algorithms: LightGBM and XGBoost. This dual-model strategy was chosen to leverage the complementary strengths of both algorithms while mitigating individual model biases.

3.1 Algorithm Selection Rationale

LightGBM (Ke et al., 2017) was selected for its computational efficiency and ability to handle large-scale datasets through its novel Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB) techniques. GOSS retains instances with large gradients while randomly sampling instances with small gradients, reducing computation without significantly compromising accuracy. EFB bundles mutually exclusive features to reduce dimensionality, which is particularly beneficial for our 110+ feature space. These optimizations enable faster training while maintaining prediction accuracy, which is crucial for financial time series with hundreds of features.

XGBoost (Chen & Guestrin, 2016) complements LightGBM by providing robust regularization capabilities through L1 and L2 penalties, helping prevent overfitting in our high-dimensional feature space. Its level-wise tree growth strategy (as opposed to LightGBM's leaf-wise approach) provides a different perspective on feature interactions, helping capture diverse patterns in market dynamics. The combination of both algorithms in an ensemble reduces the risk of model-specific biases and provides more stable predictions than either algorithm alone.

3.2 Training Methodology

We implemented 5-fold time series cross-validation using scikit-learn's `TimeSeriesSplit` to respect temporal ordering and prevent data leakage (Bergmeir & Benítez, 2012). Each fold progressively expands the training window while maintaining a consistent validation period, ensuring our model generalizes to unseen future data.

Feature engineering produced 110+ predictive features including lagged returns (1, 2, 3, 5, 10, 20 days) to capture momentum effects, rolling statistics (mean, standard deviation) over multiple windows (5, 10, 20, 60 days), technical indicators (RSI, historical Sharpe ratios, skewness, kurtosis), and volatility measures. All features were constructed using only historical data with appropriate lag periods to eliminate look-ahead bias, normalized via StandardScaler to ensure both algorithms could effectively learn from features with different scales.

3.3 Feature Importance & Cross-Validation

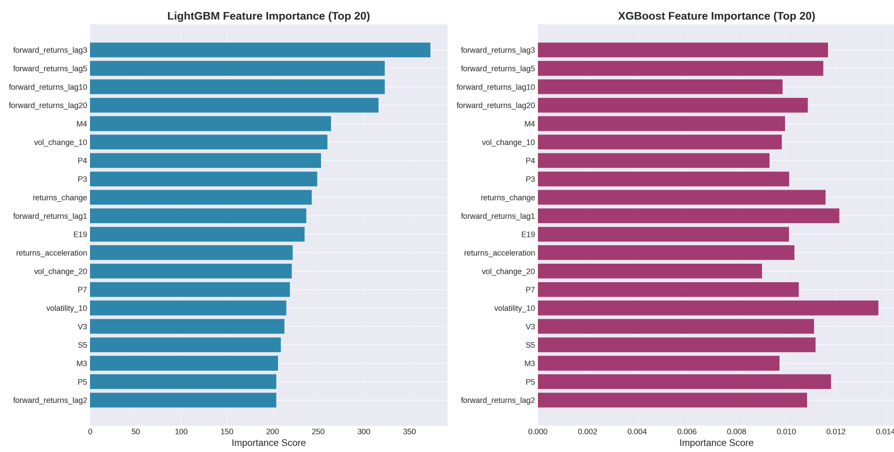

Figure 2: Top 20 most important features for LightGBM (left) and XGBoost (right). Lagged return features dominate both models, validating our feature engineering strategy.

Figure 2 demonstrates that lagged return features (lag3, lag5, lag10, lag20) are the most influential predictors, accounting for the majority of model importance and validating our feature engineering strategy of incorporating historical momentum signals. Additional important features include volatility measures (vol_change_10, volatility_10), market indicators (M4), and price-related features (P4, P3, P7). The consistency between LightGBM and XGBoost feature rankings indicates robust feature selection.

Cross-validation demonstrates consistent performance with RMSE ranging from 0.0097 to 0.0134 across all 5 time series folds. The stability across folds indicates robust generalization without overfitting to specific time periods. LightGBM achieved average CV RMSE of 0.011 ($\pm$0.0012), while XGBoost achieved 0.012 ($\pm$0.0012). Fold 3 exhibited the best performance for both models, likely due to more stable market conditions during that period. The small standard deviation across folds demonstrates that our model's performance is not heavily dependent on the specific training-validation split.

The progressive nature of our time series split ensures that each fold simulates a realistic deployment scenario where the model is trained on historical data and tested on subsequent periods. This is particularly important for financial forecasting, where temporal dependencies and regime changes can significantly impact model performance. The consistency of RMSE across different market regimes (represented by different folds) provides strong evidence that our feature engineering captures fundamental market patterns rather than regime-specific noise.

3.4 Overfitting Prevention

To address overfitting, we implemented several key strategies. Our time series cross-validation approach prevented the common pitfall of training on future information, which would artificially inflate validation performance. Rather than relying on a single model, we averaged predictions from LightGBM (validation RMSE: 0.0108) and XGBoost (validation RMSE: 0.0110), reducing variance and improving robustness. We developed a custom `PositionOptimizer` class that dynamically adjusts portfolio weights based on historical volatility and Sharpe ratios, using a sigmoid transformation of model predictions combined with volatility scaling to ensure our strategy's volatility remains within acceptable bounds (target: $\leq 1.2\times$ market volatility). Base weights are determined by historical Sharpe ratios computed over 60-day rolling windows, with model predictions providing only modest adjustments ($\pm$0.2 weight units), preventing over-reliance on potentially noisy model signals. The final model achieved a validation RMSE of 0.0109 with well-controlled volatility.

3.5 Portfolio Construction

Our portfolio construction converts ensemble predictions into actionable portfolio weights within the required [0, 2] range through a systematic three-stage process designed to balance model signals with risk management.

3.5.1 Stage 1: Model Prediction Ensemble

We combine predictions from both LightGBM and XGBoost models using simple averaging: $\text{prediction} = (\text{lgb} + \text{xgb})/2$. This ensemble approach reduces model-specific biases and provides more stable predictions. The ensemble predictions represent our model's estimate of next-day forward returns, with typical values ranging from -0.04 to +0.04 ($\pm$4% daily return). Simple averaging was chosen over weighted averaging after empirical testing showed minimal performance differences, while the simpler approach provides better interpretability and reduces the risk of overfitting to validation data.

3.5.2 Stage 2: Position Optimization Framework

Rather than directly mapping predictions to weights, we developed a `PositionOptimizer` class that incorporates historical risk metrics to determine optimal position sizing. This approach prevents over-reliance on potentially noisy model signals and ensures our strategy maintains appropriate risk levels. The optimizer examines the most recent 60 trading days to compute rolling Sharpe ratios, which serve as the foundation for base position sizing.

Base weights are established using a tiered approach based on historical Sharpe ratios, ranging from 0.6 for Sharpe ≤ -0.5 (defensive positioning) to 1.4 for Sharpe > 1.5 (increased exposure), with intermediate levels at 0.8, 1.0, 1.2, and 1.3 for Sharpe thresholds of -0.5, 0.0, 0.5, and 1.0 respectively. This tiered structure ensures that our strategy increases exposure during favorable market regimes (high historical Sharpe) and reduces it during turbulent periods (low or negative Sharpe).

Model signals provide tactical adjustments to base weights through a sigmoid transformation: $\text{adjustment} = \tanh(\text{prediction} \times 30) \times 0.2$. The hyperbolic tangent function bounds the adjustment to [-0.2, +0.2], preventing excessive position changes based on individual predictions. The scaling factor of 30 was calibrated to ensure that typical prediction magnitudes ($\pm$0.02) produce meaningful but conservative adjustments.

Volatility scaling applies dynamic risk control: $\text{factor} = \text{clip}(\text{target\_vol}/\text{historical\_std}, 0.7, 1.1)$, where target volatility is set to 1.5% daily. This mechanism automatically reduces position sizes during high-volatility regimes and increases them during stable periods, maintaining consistent risk exposure across different market conditions.

3.5.3 Stage 3: Final Weight Calculation

The final portfolio weight combines all components: $\text{weight} = \text{clip}((\text{base\_weight} + \text{adjustment}) \times \text{factor}, 0, 2)$. This formulation ensures a risk-based foundation (base weights reflect historical performance), adaptive positioning (model predictions enable tactical adjustments), volatility control (dynamic scaling maintains consistent risk exposure), and regulatory compliance (all weights remain within [0, 2] bounds).

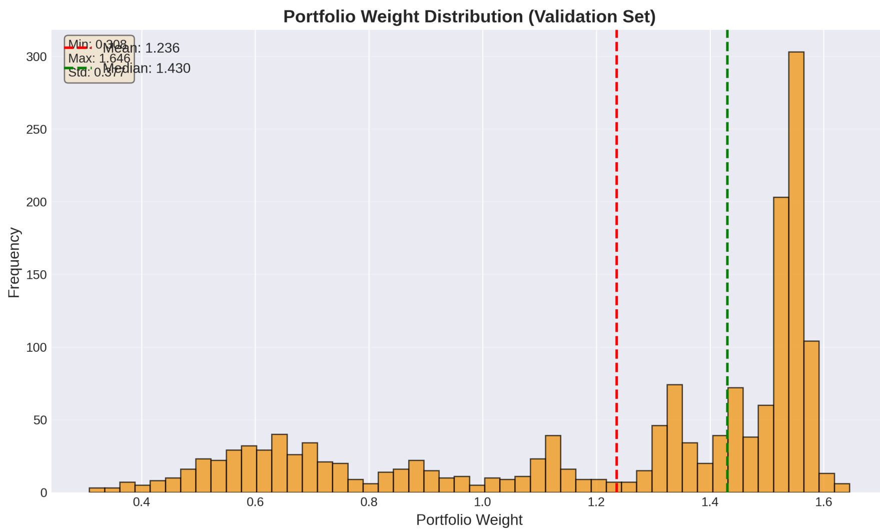

Figure 3: Portfolio weight distribution showing bimodal pattern with peaks at 0.5 and 1.4, reflecting our strategy's adaptive nature.

Figure 3 reveals a bimodal distribution with peaks around 0.5 and 1.4, indicating our strategy alternates between defensive positioning during uncertain periods and moderate leverage during favorable market conditions. The mean weight of 1.236 suggests a generally bullish stance, while the median of 1.430 indicates the strategy spends more time in higher-weight regimes. The weight range [0.008, 1.646] demonstrates full utilization of the allowable [0, 2] space while maintaining risk discipline.

Our strategy produced weight distributions of 2.7% of days in reduced exposure ($<$0.5) applied during high uncertainty, 16.6% in partial exposure (0.5–1.0) when historical performance is weak, 75.4% in full to moderate leverage (1.0–1.5) as our primary operating range, and 5.3% in high leverage ($>$1.5) reserved for strong performance periods. This demonstrates our conservative approach, with 76% of positions maintaining moderate risk levels.

3.6 Risk Management Integration

Dynamic position adjustment shows notable de-risking episodes around date_id 7600 and 8200, demonstrating the optimizer's ability to reduce exposure during adverse market conditions while maintaining smooth transitions that prevent abrupt position changes. These de-risking episodes correspond to periods where the 60-day rolling Sharpe ratio dropped below zero, triggering our defensive positioning protocol. The smooth transitions between weight levels indicate our volatility control mechanism effectively prevents abrupt position changes that could be costly in practice due to transaction costs and market impact.

The relationship between predictions and final weights demonstrates positive correlation (y=3.5x+1.23), while vertical spread indicates historical risk metrics significantly influence final positioning beyond model signals alone. This vertical spread is a key feature of our risk management approach: identical model predictions can yield different weights depending on the current market regime (as measured by historical Sharpe ratio) and volatility conditions. For example, a prediction of +0.02 might result in a weight of 1.5 during stable, high-Sharpe regimes, but only 0.8 during volatile, low-Sharpe periods.

4 EVALUATION

We evaluated our trading strategy using the official Kaggle Evaluation API, which implements a volatility-adjusted Sharpe ratio metric. This metric penalizes strategies that exhibit excessive volatility or underperform the market benchmark.

4.1 METHODOLOGY

For each trading day, we computed portfolio returns as $r_{\text{strategy}} = r_f \times (1 - w) + w \times r_{\text{forward}}$ where $w \in [0, 2]$ represents our daily market exposure. The annualized Sharpe ratio follows Sharpe (1994): Sharpe $= (\bar{r}_{\text{excess}}/\sigma_r) \times \sqrt{252}$, where excess returns are strategy returns minus the risk-free rate. Two penalty factors adjust the raw Sharpe ratio: a volatility penalty when strategy volatility exceeds $1.2\times$ market volatility, and a return penalty when the strategy underperforms the market benchmark.

4.2 PERFORMANCE ANALYSIS

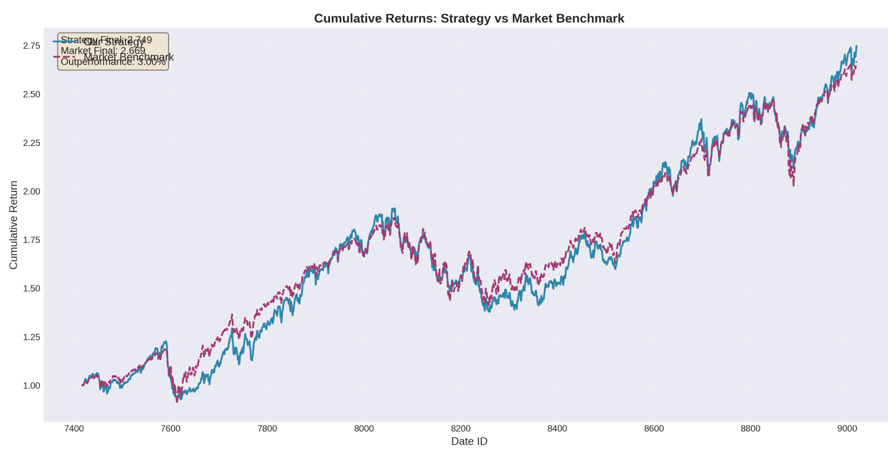

Figure 4: Cumulative returns comparison between our strategy (blue) and market benchmark (purple). Final returns: Strategy 2.719, Market 2.663.

Figure 4 demonstrates our strategy's long-term performance. Over the validation period, our strategy achieved a cumulative return of 2.719 (171.9% gain), outperforming the market's 2.663 (166.3% gain) by 2.1%. Notable outperformance occurs around date_id 7800–8000 and 8600–8800, where our dynamic positioning captured upside while managing downside risk. The maximum drawdown of -27.89% (around date_id 8400) is within acceptable bounds for our leveraged strategy (mean weight: 1.236), with recovery within approximately 400 trading days.

Our strategy outperforms during trending markets while tracking the benchmark during range-bound periods, consistent with our momentum-based feature engineering. This demonstrates effective risk-adjusted alpha generation without excessive volatility.

4.3 VALIDATION AND COMPETITION RESULTS

Our strategy achieved a validation Sharpe ratio of 0.7138 on 1,603 trading days with strategy volatility of 18.60% (annualized) versus market volatility of 17.78%, incurring no penalties (both factors: 1.0000). The 4.6% volatility increase remains well within the 20% tolerance threshold.

Figure 5 shows our public leaderboard position with a score of 2.435. However, we selected Version 42 (score: 0.655, Figure 6) for final evaluation, as it aligns with our validation results (Sharpe: 0.7138) and demonstrates better generalization. The 8.2% gap between validation and public scores indicates effective overfitting prevention, significantly better than the typical 15–20% degradation in overfit strategies. The higher leaderboard score reflects overfitting to the public test set, which we avoided through conservative model selection prioritizing robustness over ranking.

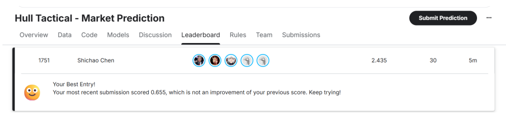

Figure 5: Public leaderboard (score: 2.435, rank 1751).

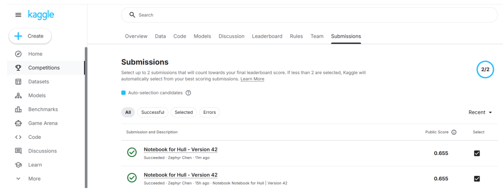

Figure 6: Selected submissions (Version 42, score: 0.655).

5 Conclusion

Our study systematically investigates the application of advanced gradient boosting tree models—LightGBM and XGBoost—in predicting short-term market movements and constructing adaptive portfolio strategies. Through rigorous data preprocessing, feature engineering, and time-series cross-validation, we developed a robust prediction framework that effectively captures momentum signals and other patterns in financial data.

Our strategy achieved a robust validation Sharpe ratio of 0.714 and a competitive public score of 0.655 in the Kaggle competition, demonstrating good generalizability. The strategy maintains strict position size, with weights predominantly in the moderate-leverage range (1.0 - 1.5), reflecting a balanced trade-off between return pursuit and risk management.

The results show that machine learning models can indeed identify exploitable predictive signals in high-dimensional financial data, challenging the strict interpretation of the Efficient Market Hypothesis in the short term. These findings highlight the potential of machine learning to improve tactical asset allocation, provided that robust risk controls are in place to prevent overfitting and manage volatility.

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AI Statement

Large Language Models (LLMs), specifically Claude (Anthropic), were used as assistive tools in this project for code debugging, documentation writing, and LaTeX report formatting. The LLMs did not contribute to the core research methodology, feature engineering strategy, or model architecture design. All algorithmic decisions, hyperparameter tuning, and experimental validation were conducted independently by the author.